

CHENNAI INTL, India

WMO: 432790

Lat: 12.994N

Lon: 80.181E

Elev: 52

StdP: 14.67

Time Zone: 5.50 (IND)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	68.0	69.6	58.7	74.0	84.0	61.1	80.5	82.2	17.6	83.2	15.7	83.4	0.9	320	0.376

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	16.6	102.2	78.8	100.2	79.1	98.3	78.9	83.4	91.9	82.6	90.6	81.9	89.8	9.0	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
81.0	161.4	87.3	80.7	159.7	87.1	80.2	156.9	86.8	47.5	92.6	46.6	90.8	46.0	89.5	89.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
17.7	15.6	13.9	DB	64.9	107.1	2.1	2.5	63.4	108.9	62.2	110.4	61.0	111.8	59.5	113.6	
			WB	61.7	85.4	2.2	1.2	60.1	86.3	58.8	87.0	57.6	87.7	55.9	88.6	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	84.3	77.7	79.9	83.7	87.8	91.1	89.9	87.7	86.3	85.6	83.1	80.1	78.0
	DBStd	5.11	1.87	2.28	2.62	2.70	3.46	3.31	2.98	2.59	2.34	3.02	2.31	2.05
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD50	12509	860	837	1046	1134	1273	1197	1168	1126	1069	1027	903	869
	CDD65	7034	395	417	581	684	808	747	703	661	619	562	453	404
	CDH74	83640	2965	3873	6712	9133	11439	10543	9523	8502	7719	6224	4015	2992
	CDH80	40602	770	1362	2996	4903	7020	6292	5189	4244	3638	2425	1097	666
(14) Wind	WSAvg	6.3	4.6	5.1	6.1	7.3	8.4	8.7	8.0	7.5	6.2	4.5	4.3	4.9
(15) Precipitation	PrecAvg	54.3	0.9	0.2	0.3	0.6	1.2	3.1	4.6	5.9	5.6	11.2	14.2	6.3
	PrecMax	87.2	6.7	2.0	6.9	4.5	10.0	20.2	8.6	15.6	13.5	31.5	42.2	27.2
	PrecMin	27.9	0.0	0.0	0.0	0.0	0.0	0.3	0.7	1.0	0.8	0.1	0.5	0.0
	PrecStd	11.9	1.5	0.4	0.9	1.0	1.7	2.8	2.1	3.2	3.0	6.3	9.3	5.9
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	88.2	93.0	97.1	101.1	106.3	104.1	100.7	98.8	97.1	95.2	91.3	88.1
		MCWB	73.7	74.8	76.6	80.5	78.7	78.9	77.7	78.5	78.7	78.1	77.5	75.9
	2%	DB	86.4	89.9	94.8	98.3	103.7	101.1	98.5	96.7	95.1	93.2	88.2	86.2
		MCWB	73.5	74.3	77.0	80.5	78.7	78.3	77.6	78.3	78.4	77.9	76.7	74.9
	5%	DB	85.7	88.1	92.9	96.4	100.7	98.9	96.7	94.9	93.4	91.3	87.4	84.6
		MCWB	73.2	74.1	77.2	80.2	79.9	78.0	77.8	78.2	78.3	78.2	76.5	73.9
	10%	DB	84.2	86.9	91.0	94.4	98.4	97.0	94.8	93.0	91.7	89.5	85.8	83.9
		MCWB	72.3	73.8	76.8	79.9	80.0	78.0	77.6	78.2	78.3	78.1	76.0	73.5
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	77.8	79.2	81.6	84.2	84.7	83.5	82.0	82.5	82.5	82.1	80.8	79.1
		MCDB	83.7	85.9	89.3	95.5	94.2	93.1	89.1	89.9	89.3	87.6	85.6	83.3
	2%	WB	76.5	78.0	80.3	82.7	83.7	82.2	80.9	81.3	81.3	81.1	79.7	78.1
		MCDB	82.5	84.9	87.6	90.7	92.3	90.7	88.8	89.1	88.1	86.9	84.4	82.3
	5%	WB	75.5	77.0	79.6	81.9	82.8	81.2	80.1	80.5	80.6	80.2	78.9	77.1
		MCDB	81.5	84.4	87.0	89.3	91.4	90.3	88.6	88.4	87.4	86.2	83.4	81.5
	10%	WB	74.5	76.1	78.9	81.3	82.2	80.5	79.5	79.8	79.9	79.6	78.2	76.2
		MCDB	80.5	83.2	86.4	89.0	90.8	89.9	88.5	87.8	86.9	85.5	82.6	80.6
(35) Mean Daily Temperature Range	5% DB	MDBR	14.2	15.7	16.0	14.8	16.6	15.8	14.6	14.2	13.8	12.1	11.0	12.0
		MCDBR	15.4	16.9	17.5	16.5	19.4	17.6	16.7	16.2	15.6	14.8	13.4	13.4
		MCWBR	5.5	5.5	5.5	5.1	6.2	5.9	5.7	5.5	5.4	5.2	5.1	5.3
	5% WB	MCDBR	13.1	14.5	15.5	15.1	16.8	16.3	15.2	14.6	14.2	12.5	10.5	10.5
		MCWBR	5.6	5.2	5.2	5.2	5.9	6.4	5.9	5.7	5.6	5.3	5.1	5.3
(40) Clear-Sky Solar Irradiance	taub		0.468	0.458	0.504	0.522	0.601	0.538	0.535	0.524	0.519	0.533	0.511	0.516
	taud		2.160	2.195	2.074	2.073	1.884	2.077	2.087	2.125	2.139	2.088	2.112	2.050
	Ebn at Noon		263	272	261	255	232	244	245	251	254	247	249	245
	Edh at Noon		47	47	54	54	64	52	52	51	50	52	49	51
(44) All-Sky Solar Radiation	RadAvg		1576	1858	2009	2058	1900	1697	1590	1610	1646	1472	1333	1331
	RadStd		52	60	70	71	95	117	121	86	102	91	126	115

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	+0.50	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (2 neighbors)	+0.40	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page